Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Report

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PHYS 400 — Final Presentation

Outline

Why Two Elastic Constants Govern Structural Design

Every load-bearing part is sized by two numbers:

- $E = \sigma/\varepsilon$ — Young's modulus: stiffness (GPa).
- $\nu = -\varepsilon_{\mathrm{lat}}/\varepsilon_{\mathrm{axial}}$ — Poisson's ratio: transverse contraction.

For cubic crystals both follow from the two stiffness constants C_{11} and C_{12} — **the quantities this work targets from first principles**.

They set: beam deflection $\delta \propto 1/E$, buckling load $\propto E$, natural frequency $\propto \sqrt{E/\rho}$; bulk modulus $K = E/[3(1-2\nu)]$, stress concentration, toughness.

Stability (cubic, Born–Cauchy): $C_{11} > C_{12}$.

Representative structural applications

Application	Material	E (GPa)	ν
Aerospace fuselage	Al Alloys	60–80	0.3–0.38
Turbine disk	Inconel 718	200	0.29
Orthopaedic implant	Ti 6Al–4V	110	0.34
Automotive structure	Mg AZ91	45	0.35
Structural steel	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28

Same two constants span polymers ($\nu \to 0.4$) to steel — and fix whether a design survives load.

Predicting C_{11}, C_{12} from Atoms: Why We Need Potentials and DFT

Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{\mathrm{r}_i\})$; forces obtained as $\mathrm{F}_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard–Jones, Morse) — too crude for metals.
- Embedded Atom Method (1984) — good for FCC metals.
- **Modified EAM** (MEAM) [1]: adds angular terms — covers BCC, HCP and mixed bonding.

Density-functional theory (DFT). First-principles electronic-structure method (Kohn–Sham equations [2]); here a plane-wave + pseudopotential basis in Quantum Espresso [3].

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0, bulk modulus B
- Binary formation energies $E_{\mathrm{form}}(i, j)$
- Single-crystal elastic constants C_{11}, C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).
MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).
DFT provides the reference dataset against which the MEAM potential is calibrated.

Why No Existing Potential Suffices: The MEAM Coverage Gap

Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by Al 7075.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [4–6]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.

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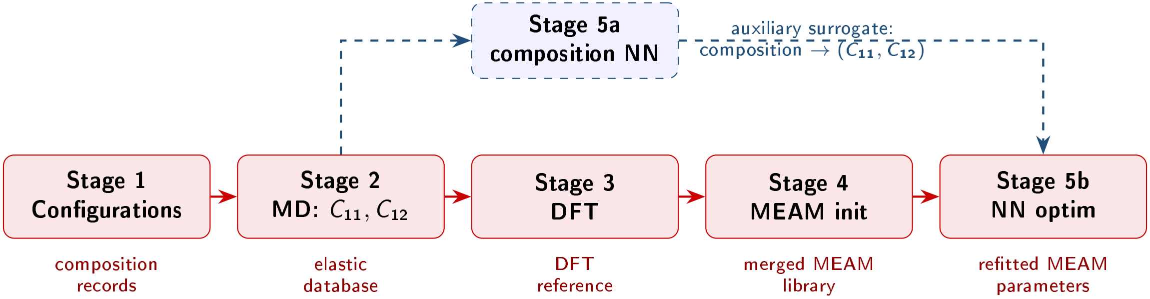
Method — five-stage pipeline

Proposed Methodology: DFT-Informed NN Optimisation



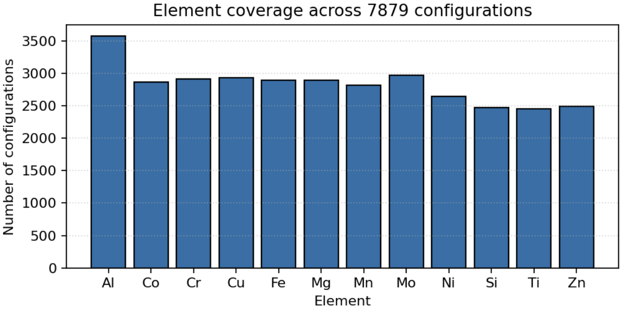
The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM → (C_{11}, C_{12}) map by gradient descent, with single-element parameters anchored to DFT references.

MD = Molecular Dynamics; DFT = Density-Functional Theory; NN = Neural Network.

Stage 1: Alloy Configuration Generation

Sampling.
Uniform draws on the 12-element composition simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);
Al-weighted to oversample the aluminium-alloy region.

Corpus statistics.
∼7900 configurations generated;
7815 retained after the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).



Element-occurrence distribution over the retained corpus.

LAMMPS

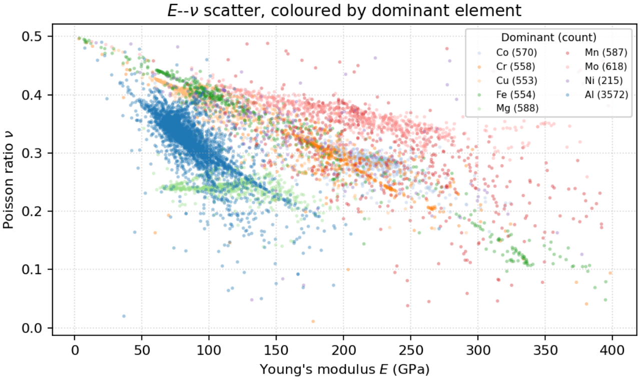
Implementation. [7]
LAMMPS molecular dynamics with the merged 12-element MEAM potential.

Per-configuration protocol:

- $\sim$4 nm cubic supercell
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\rightarrow C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.



E–ν distribution, coloured by dominant element.

Stage 3–4: DFT References → MEAM Initialisation

Quantum Espresso

DFT references + existing MEAM-file cross-interactions → **MEAM initialisation** α + mixing sign → **12-element MEAM library**

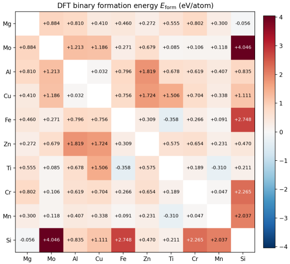
Single-element terms

Each element's $(a_0, E_{\mathrm{coh}}, B)$ pins the universal binding curve — spacing, well depth, curvature — fixing the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{\mathrm{coh}}} \qquad (\Omega : \text{atomic volume}).$$

Scope. 12 elements (FCC/BCC/HCP/diamond); $\alpha \approx 4.5$ (Al, Si) to 10.9 (Zn). PBE–GGA plane waves (Quantum Espresso); a_0, B from an equation-of-state fit.

Cross-terms



Binary formation energies $E_{\mathrm{form}}(i, j)$ (eV/atom): negative = exothermic mixing, positive = limited solubility — the sign seeds each cross-term.

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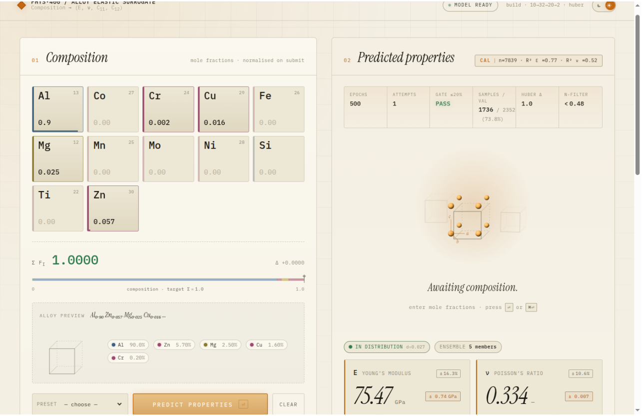
Results

Architecture. $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$ fully-connected MLP
with ReLU activation; Huber loss on $\hat{C}_{11}, \hat{C}_{12}$;
deep ensemble of five seeds [8] for epistemic uncertainty.

Result

Validation Set (30 % of the dataset): $R^2_E = 0.77$, $R^2_\nu = 0.52$.

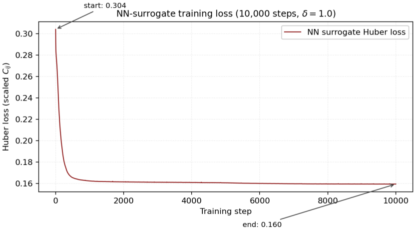
Al 7075 Example: $\hat{E} = 75.5$ GPa, $\hat{\nu} = 0.334$ (published: 71–72 GPa, $\nu \approx 0.33$ — within ~ 5 %).



Interactive composition predictor.

Closed-loop training process

1. **Surrogate.** A 4-layer MLP ($\to 20 \to 20 \to 10 \to$, ReLU) learns (MEAM params) $\to$ (C_{11}, C_{12}).
2. **Fit.** Trained with *Adam* [9] on a *Huber* loss ($\delta = 1$), up to 10^4 epochs, early-stopped once the loss falls below 5×10^{-3}.
3. **Inverse step.** Adam (lr 0.01, 3000 steps) descends through the *frozen* surrogate to the MEAM parameters that hit the target (C_{11}, C_{12}).
4. **Loop.** Optimised parameters are re-evaluated in LAMMPS; new samples are appended and the surrogate retrained (active learning).



Surrogate Huber-loss decays 1.0 $\to$ 0.16 over training.

Why these choices

Adam — adaptive-moment gradient optimiser: per-parameter step sizes from running gradient moments, a robust default for NN training.
Huber loss — squared-error for small residuals, absolute-error for large ones; outlier configurations (Mg/Zn) cannot dominate the fit.

Result — a working proof of concept

Tested on **30 unseen alloys**:

- **11** gave a physically stable potential;
- **4** matched the MD reference within $\sim$20 %.

The pipeline runs end-to-end; accuracy grows with more data and longer training (next slide).

External Validation: Quantitative Summary

By material family ($N = 43$)

		E MAPE (%)		ν MAPE (%)	
Family	N	ML	MEAM	ML	MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Ni–Fe (A-286)	1	107.0	—	—	—
Overall	43	**10.8**	42.2	**7.2**	7.1

Aluminium by AA series ($N = 33$)

		$\langle E_{\mathrm{lit}} \rangle$	E MAPE (%)	
Al series	N	(GPa)	ML	MEAM
2xxx (Al–Cu)	11	72.7	1.3	8.0
5xxx (Al–Mg)	9	70.5	2.1	7.9
6xxx (Al–Mg–Si)	7	68.9	3.8	4.2
7xxx (Al–Zn–Mg)	6	71.8	2.8	9.3
All aluminium	33	71.1	**2.3**	7.4

Both surrogates reproduce aluminium Young's modulus to within a few percent (ML 2.3 %, MEAM 7.4 %) across all four alloy series; error grows only *off* the Al-rich design domain — steeply for MEAM, gently for the composition NN.

43 basis-covered alloys; MAPE vs experimental data from ASM/MatWeb [10]. Unphysical MEAM runs excluded.

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Discussion & summary

Summary & Further Studies

Achieved

- ✓ **Automated 12-element MEAM pipeline** — an end-to-end workflow that assembles a multi-element potential.
- ✓ **Aluminium-alloy elastic prediction** — E, ν and C_{11}, C_{12} obtained from composition.
- ✓ **DFT-seeded, NN-trained MEAM** — first-principles references refit through a neural surrogate.

What comes next

1. **More data coverage** — wider sampling across the composition simplex and crystal structures.
2. **More NN training epochs** — deeper surrogate convergence before each refit.
3. **Denser atomic-cell DFT** — finer k-point and supercell sampling for sharper references.

The architecture is established — the open frontier is *data*, *training depth*, and *DFT resolution*.

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